

DÁIL ÉIREANN

AN COMHCHOISTE UM OIDEACHAS, BREISOIDEACHAS AGUS ÁRDOIDEACHAS, TAIGHDE, NUÁLAÍOCHT AGUS EOLAÍOCHT

JOINT COMMITTEE ON EDUCATION, FURTHER AND HIGHER EDUCATION, RESEARCH, INNOVATION AND SCIENCE

Dé Máirt, 18 Aibreán 2023

Tuesday, 18 April 2023

Tháinig an Comhchoiste le chéile ag 11 a.m.

The Joint Committee met at 11 a.m.

Comhaltaí a bhí i láthair / Members present:

Teachtaí Dála / Deputies	Seanadóirí / Senators
Sorca Clarke,*	Eileen Flynn,
Mairéad Farrell,*	Rónán Mullen,
Jim O'Callaghan.	Fiona O'Loughlin,
	Pauline O'Reilly.

* In éagmais / In the absence of Deputies Rose Conway-Walsh and Donnchadh Ó Laoghaire.

I láthair / In attendance: Deputy Mattie McGrath and Senators Malcolm Byrne and Maria Byrne.

Teachta / Deputy Paul Kehoe sa Chathaoir / in the Chair.

The Future of STEM in Irish Education: Discussion (Resumed)

An Cathaoirleach: I remind members to ensure their mobile phones are switched off for the duration of the meeting as they interfere with the broadcasting equipment, even when on silent mode. Are the minutes of the meeting of 28 March agreed? Agreed.

Before I proceed - I should have done this in private session - I welcome the new Sinn Féin spokespersons, Deputy Sorca Clarke, spokesperson on education, and Deputy Mairéad Farrell, spokesperson on higher education, both of whom will be nominated as members of the committee. I thank Deputies Rose Conway-Walsh and Donnchadh Ó Laoghaire for their input to the committee over recent years. They were two very good members of the committee.

We will have two sessions today. On behalf of the committee, I welcome Dr. Cornelia Connolly, associate professor at the school of education, College of Arts, Social Sciences and Celtic Studies, University of Galway; Dr. Ann Marcus-Quinn, lecturer in technical communication and instructional design, University of Limerick; Professor Sarah McCormack, professor in energy engineering, department of civil, structural and environmental engineering, Trinity College Dublin; Professor Hamsa Venkat, professor of STEM education for primary and early childhood education, Institute of Education, Dublin City University, DCU; Dr. Margaret Leahy, head of the school of STEM education, innovation and global studies, Institute of Education, DCU; and Dr. Áine Hyland, emeritus professor of education, University College Cork, UCC.

The witnesses are here to discuss the future of STEM in Irish education. The format of the meeting is that I will invite the witnesses to make brief opening statements in the following order: Dr. Connolly, Dr. Marcus-Quinn, Professor McCormack, Professor Venkat, Dr. Leahy and Professor Hyland. This will be followed by questions from members of the committee. Each member has a five-minute slot, given the time constraints because we have two sessions, and that is to ask their questions and for the witnesses to respond. The committee will publish the opening statements on its website following today's meeting.

I remind members of the long-standing parliamentary practice to the effect that they should not comment on, criticise or make charges against a person outside the Houses or an official either by name or in such a way as to make him or her identifiable. Witnesses are reminded of the long-standing parliamentary practice that they should not criticise or make charges against any person or entity by name or in such a way as to make him, her or it identifiable or otherwise engage in speech that might be regarded as damaging to the good name of a person or entity. Therefore, if their statements are potentially defamatory in respect of an identifiable person or entity, they will be directed by me, as Chair, to discontinue their remarks, and it is imperative that they comply with any such direction.

Dr. Connolly may begin, and the other witnesses will follow. The witnesses have five minutes each.

Dr. Cornelia Connolly: Dia dhaoibh, a Chathaoirligh agus a dhaoine uaisle. Gabhaim búiochas leis an gcoiste as ucht an chuiridh a bheith anseo agus caint libh faoi thodhchaí eolaíochta, teicneolaíochta, innealtóireachta agus mata i gcóras oideachais na hÉireann. I am a computer engineering graduate, an associate professor at the school of education in University of Galway and a funded investigator with Lero, the Science Foundation Ireland, SFI, research centre for software. I state this merely because the terms “software”, “IT” and “computing” do not fall within the STEM acronym, yet coding and computational thinking, constituent ele-

ments of computer science, are fundamental and underpin all of science, technology, engineering and mathematics.

Computer science is a key discipline necessary for our future and societal development, as outlined in my written submission. In considering the disposition of young people when completing their formal education, the key competencies necessary for their future, I will emphasise three points about computer science education and its role in the future of STEM in Irish education.

Ireland is working to become a digital leader at the heart of European and global development. The recently published digital Ireland framework sets out a roadmap for the ongoing digital transformation of the economy and the need to strengthen the centrality of education, research and innovation. Digital skills and a flourishing computer science education ecosystem are essential to our national digital transformation. To grow our digital economy, Ireland needs an advanced workforce ready to take advantage of the opportunities that transformation will bring. Opportunities to learn digital skills must be provided from an early age. That includes computer science education, along with comprehensive knowledge and understanding of data-intensive technologies, such as artificial intelligence, AI. Computer science, CS, is an important part of STEM. CS is the study of computing technology, how coding, programming and computational thinking can be used to solve problems and how computing technology impacts the world around us. The knowledge, ways of thinking, problem-solving and creativity involved in the diverse field of CS are invaluable skills which bring the benefits of innovation and digital transformation to everyone in society.

In recent years, the Irish education system has embraced CS by bringing it into the curriculum. Nevertheless, we are a long way off making this important subject available to all students. Equity of access is a matter of concern. Currently, CS at senior cycle is limited by the low number of post-primary schools offering coding at junior cycle and by a significant shortage of teachers qualified to teach the subject. The capacity of the Irish education system to facilitate CS education is highly contested, with constraints being placed on school timetables. Coupled with this is the low level of awareness and a pervading misunderstanding of what the discipline is and what it is not. For example, digital literacy, on the one hand, would allow a Deputy or Senator to write his or her election manifesto and circulate it via email; computer science, on the other hand, would allow him or her to use an algorithm to monitor voting patterns or constituents' concerns.

The Programme for International Student Assessment, PISA, review on education will examine informatics, as the subject is commonly known in Europe, in 2024, which will include the study of the data, structure and computational systems. As a result, the new primary curriculum framework offers a promising opportunity for the introduction of CS to all schoolchildren in Ireland. Professional development in CS education should no longer be viewed as an elective for primary teachers but, rather, embedded within all initial teacher education at primary and post-primary.

We need to develop a shared understanding and strengthen the acceptance of CS as a foundational competence for every child, irrespective of race, gender or socioeconomic background, enabling the young people of today to become active participants in the digital society of tomorrow and to become the designers and developers, not just the consumers, of technology. Broad engagement is needed with key stakeholders across the education system and with industry partners in an appropriate manner to highlight and optimise the benefits of CS as a key competency. We also need a holistic approach to the introduction of CS in formal education and to

create a learning pathway from preschool to primary to post-primary. To ensure this pathway, CS must be embedded within initial teacher education and we need continued investment in in-service teacher programmes.

In preparing the groundbreaking 1965 Investment in Education report, one of the distinguished authors, Bill Hyland, used CS in analysing the use of resources in schools. That seminal report was informed by the concept of education as a means by which we invest in the next generation. That mission remains of importance today, particularly in the context of CS education.

Dr. Ann Marcus-Quinn: I thank the committee for the opportunity to speak on the future of STEM education in the context of access and equity in the current digital strategy.

Schools, particularly post-primary schools, have changed remarkably since March 2020. The inspectorate's Looking at Our School 2022 acknowledges this in its quality framework. In February 2020, a review of tablet devices and how they were used in one school was formally published. There was no opportunity to discuss the findings and recommendations of this review because the country was in the grip of a pandemic. We are now in the grip of a cost-of-living crisis. More schools require students to purchase a mandated device to participate in classroom activities and complete parts of the State examinations. We need wider societal dialogue about how technology is embedded in our schools.

The use of ICT has been embedded into all six key skills for junior cycle. This is progressive, especially as technology becomes more and more embedded in many aspects of our daily lives. The digital strategy for post-primary is ambitious and provides a comprehensive roadmap that endeavours to ensure that our school system is preparing our school communities for the modern world.

International research recognises the need for appropriate investment in technology in education. In the final report of the expert group on quality education for the European Commission, the authors recommended that significant policy attention should be given to and investment should be made in improving the digital infrastructure in schools. This involves hardware, technical support and software. The Department of Education has increased the level of investment in the digital strategy for schools in recent years. However, given the level of funding available and the autonomy of schools, there remains a great deal of inequality in terms of how technologies are used to provide a quality and equitable education for all.

The three most common scenarios in schools are as follows. In scenario A, the school is a one-to-one device school. Every student must purchase a specific device for classroom work. Some 23% of parents responding to the 2022 Barnardos survey stated that they had to pay more than €300 for digital technology for their child. For many parents, the cost is closer to €1,000 when other costs are factored in. This scenario puts a financial burden on parents that varies widely depending on the school. Not all one-to-one device schools are using the same quality of digital content. In some schools, teachers cannot use textbooks or ebooks. Teachers may have to develop their own digital resources. This expectation of such content development is leading to several problems, including copyright concerns and an undue burden on teachers to become instructional designers.

In scenario B, the school has a bring-your-own-device, BYOD, policy. The range of devices being used varies hugely. The school has little control over what social media is accessed during the school day. It is much more difficult to implement any kind of digital policy in this

scenario. The risk of child safeguarding issues is much higher. This scenario essentially forces some families to buy their 12-year-olds a smartphone. Research, which includes research carried out by the ESRI, provides evidence that children under 13 with a smartphone are negatively impacted. There is also a financial cost to parents. In scenario C, the school has appropriately equipped ICT labs or an adequate trolley system in place or both for the use of technology in class. Everything is monitored and safer. No personal devices are used. This type of scenario requires significant funding, however. The new digital strategy is our best roadmap to date in terms of building digital skills and competencies. Devices are an attractive option for many schools, as having a tablet or a laptop can negate the problematic issues that can arise with smartphones. However, changing the current systems to improve the value of these purchases for both teachers and students at post-primary level is critical if equity of the digital experience is to be achieved. The Department should consider a centralised approach to both the procurement and provision of digital devices and the necessary ICT systems in order to support well-informed digital school policies and investments. The Department should also provide a recommendation regarding a common minimum standard for both student and teacher devices. A circular was published in 2017 providing recommendations on uniform purchases. A similar approach is necessary for the purchase of technology and associated services until a centralised approach and formal tendering process can be devised.

Professor Sarah McCormack: I am a professor of sustainable energy in the department of civil, structural and environmental engineering at Trinity College Dublin. I thank the committee for the invitation. Given the time restrictions and breadth of perspectives at this meeting, I will concentrate on female participation in STEM subjects. Engineering is a field that has traditionally been dominated by men. Gender diversity is not just about achieving equality; it is about harnessing the full potential of our workforce and driving innovation and progress. Women have historically faced systemic barriers and biases that have limited their participation in STEM fields. This has resulted in a significant gender gap, with women under-represented. By actively encouraging and supporting women to pursue careers in engineering and STEM areas, we can break down these barriers and promote gender equality in the workplace. It is crucial to create a level playing field where women have the same opportunities as men to enter, excel and to lead.

Encouraging women into STEM is crucial for economic growth and competitiveness. Engineering is a field that drives economic development through technological advancements and infrastructure improvements. By tapping into the talent pool of women, who make up half of our population, we can expand the workforce and drive economic growth. Studies have shown that companies with a diverse workforce, including gender diversity, tend to perform better financially and have a competitive advantage. Therefore, fostering gender diversity in engineering is not only the right thing to do, it is also smart business. From a personal perspective, I took a long route to reach my current position. Not all secondary level subjects were encouraged for girls, and being from Donegal there were not many options for third level education. Like many others, I left home to study. I went to Ulster University and studied science before switching to engineering to study for a PhD in Belfast. Afterwards, I moved to the school of physics at Technological University Dublin and then on to a faculty position in energy engineering at the school of engineering in Trinity College Dublin.

I teach more than 200 second-year students, up to 60 master's students and five PhD students. In our undergraduate engineering courses at Trinity, women make up between 20% and 30%, which is higher than most universities in Ireland. Significantly, in our engineering courses women tend to do better. They achieve higher degree results whereby more women

achieve first class degrees and women also have a higher completion rate. In 2017 it was 64% for men and more than 90% for women. Females, just like their male counterparts, have the aptitude and potential to excel in STEM fields but they often face societal and cultural barriers that discourage them from pursuing these subjects. Stereotypes and biases that suggest STEM is for boys and that girls are not as capable in these areas can limit their interest and participation. By actively encouraging girls to pursue STEM subjects in secondary schools, we can create a strong pipeline of female students who are prepared to pursue STEM careers. This will close the gender gap and ensure that girls have the same opportunities as boys to explore, learn and excel in these fields. School management, subject teachers and guidance counsellors in post-primary school settings all play pivotal roles in this regard. Encouraging STEM should be an integral part of all school culture. By equipping girls with strong STEM skills early on, we can ensure that they are prepared for the jobs of the future and can contribute to Ireland's competitiveness in industries such as technology, environment, engineering and healthcare. Companies with diverse workforces tend to perform better financially and lead to increased innovation, creativity and problem-solving. Diverse teams are better equipped to understand and address the needs of diverse communities.

Representation matters. Having visible role models can inspire and motivate young girls and women to pursue careers in engineering and other STEM fields. Showcasing successful women STEM professionals as role models in schools, universities and workplaces can help challenge gender stereotypes and encourage more girls to consider STEM as a viable career option. Encouraging gender diversity requires not only changing mindsets but also implementing supportive policies and practices in educational institutions, workplaces and the engineering profession such as flexible work arrangements, mentorship programmes, diversity and inclusion initiatives and bias training programmes.

Women offer different viewpoints, experiences and approaches to problem-solving, which can lead to more innovative solutions and improved decision-making. In a world that is facing complex challenges such as climate change, sustainable development and technological disruptions, we need diverse perspectives in engineering in order to develop innovative sustainable solutions that address the needs of all people. I thank members for their time.

An Cathaoirleach: Everyone is remaining within their five minutes. We had a group of men in here and they rambled on for seven or eight minutes. I am not sure what that indicates. I invite Professor Venkat to make her opening statement.

Professor Hamsa Venkat: I thank the Cathaoirleach and the committee for this invitation to join them today. I hold the Naughton chair in early years and primary STEM education at Dublin City University. As a result, my input will focus on issues relating to STEM education provision and access for younger children.

In recent years, STEM education has become a high priority for governments and educational policymakers around the world for reasons that are well rehearsed. There are jobs in the STEM economy and this leads to the need to close the gender gaps in participation. There is also a need for STEM-literate citizens. However, what constitutes STEM education is still poorly understood. At one extreme, STEM is simply the full discipline of science, technology, engineering and mathematics. At the other extreme, there is a fully transdisciplinary and integrated STEM that draws on content, skills and competences from all four disciplines and often other disciplines, as needed, for effective real-world problem-solving. In our work with teachers, we take the view that effective STEM education involves both, namely, supporting strong disciplinary awareness of the STEM subjects and supporting experiences of integrated STEM.

Most input to this joint committee over the past few weeks has focused on second and third level education. I am still greeted with some surprise when I say that my focus is on STEM in early years and primary settings. This response is unsurprising given that STEM education is poorly understood in a culture where proclamations of someone not being a maths person or a science person remain common. Yet, the evidence shows consistently that if we do not build confidence and enthusiasm for STEM-related learning early, by second level, the negativity and gendered stereotypes the committee has already heard about relating to STEM have already set in. The Growing Up in Ireland surveys show that attitudes towards subjects have taken form at primary level. There is also evidence from early years education showing that good-quality STEM education is linked to significantly higher enthusiasm and motivation for science in later life and is predictive of later academic achievement in mathematics knowledge and skills. The Aistear curriculum framework for early childhood and the new primary curriculum framework both address this concern head on with the insertion of STEM as an integrated curriculum area. This is welcome but teachers' lack of confidence with STEM subjects and their unfamiliarity with integrated STEM will require concerted systemic efforts to realise the promise that STEM education holds.

I will deal with the curriculum and policy-level supports while Dr. Leahy will speak to the teacher-level supports that are also critical for good quality STEM provision. At curriculum level, early childhood settings and schools need more guidance on time allocations for mathematics, science and technology alongside time for integrated STEM projects. In primary schools, mathematics has dedicated weekly time but science, technology and engineering have a relatively small monthly allocation. Given the evidence we have of science time being whittled down in the past few years in primary schools, a weekly science allocation and at least termly incorporation of integrated STEM projects would be very useful.

There are wonderful examples of integrated STEM teaching in Ireland at early years and primary level but these are not widespread enough. Providing exemplars of integrated STEM projects that illustrate openings for children to learn through play, exploration and inquiry about the STEM disciplines, which can include modelling situations, conducting fair tests and building prototype solutions, for example, and learning about transversal skills such as digital learning and working collaboratively, will be important. The NCCA, STEM professionals, teacher development bodies, the inspectorate and unions are and should continue to be part of the development of these exemplars.

Additionally, STEM projects differ depending on whether I am a scientist, technologist, engineer or mathematician. For example, coding or designing and making might be included in some STEM projects but not in others. A variety of exemplars need to be developed and shared with teachers which illustrate the breadth of STEM working. This develops practitioners' STEM confidence and competence, including their digital competence, which are critical for good-quality STEM education. Spaces and resources for STEM teaching are also critical to ensuring the successful incorporation of STEM into early education. At DCU, our work with the NCCA comparing primary curricula has shown that many countries are now supplementing their traditional mathematics and science offerings with attention to integrated STEM learning. Almost all of the countries surveyed have chosen a hybrid model of subject teaching interleaved with integrated STEM work. This overcomes the dangers of missing out key disciplinary concepts and ensures children can see the trajectory of disciplinary ideas.

I conclude by noting that, at DCU, we are excited by the opportunities offered by the attention to STEM in the Aistear and primary mathematics frameworks. There is potential for edu-

cational innovation in Ireland around STEM that serves individual and societal needs. We look forward to contributing to high-quality access to STEM experiences for young learners through our research and teacher development activities. I thank the committee for its time.

Dr. Margaret Leahy: I will do my best to keep to the five-minute limit. I thank the Cathaoirleach and committee members for the invitation to DCU Institute of Education. We engage in teacher education across the continuum from early childhood to tertiary education, which enables us to contribute to the discussion as we have an overview of STEM education across the system as it currently is. As a starting point, I reiterate that our view is that STEM education incorporates the study of the individual disciplines of STEM while also promoting an applied, problem-solving orientation to learning across and between these disciplines, in other words, integrated STEM. To achieve this, it is critical that all our students have access to the widest range of STEM education, in disciplinary and integrated forms, from early childhood to school-leaving age. Key to this provision are teachers who not only have the potential but also a responsibility to foster students' STEM interest and identities through innovative classroom practices, as well as providing access to role models and information about the diverse careers that exist within STEM industries. This is premised on teachers having the necessary content knowledge and confidence across the STEM disciplines along with the ability to integrate the STEM disciplines to provide meaningful learning opportunities.

Despite this, we know that early childhood STEM can often be of poor quality. Early childhood educators and primary teachers' content knowledge linked to the STEM disciplines can be weak in some cases, which limits the potential of integrated STEM. At primary and post-primary levels, the emphasis is on traditional subject areas, which makes integrated STEM more difficult. At post-primary, current assessment approaches, in other words, the leaving certificate, are perceived as inhibiting the use of more innovative pedagogies and the adoption of integrated STEM. The focus on State examinations may also mask a larger problem that many teachers do not understand how to implement inquiry-based teaching and learning and some teachers are not accessing the laboratory skills or subject content knowledge required to teach inquiry-based practical lessons, which is what the science syllabi demand. The quality and level of a teacher's qualification are also paramount. Recent changes in the Teaching Council imply that science teachers no longer require three credits in all junior science syllabi.

What can be done about this? Across all levels, but especially at early childhood and primary levels, there is need for a longitudinal and coherent approach to pre- and in-service teacher education, and that should include the provision of curriculum, teaching materials and resources, including the exemplar support materials my colleagues spoke of, to support STEM teaching and learning. These models should be geared towards teaching and learning that supports inquiry, experimentation and higher order thinking and skills in the STEM areas and integrated STEM. Professional learning should be supported with programmes for monitoring take-up, student experience and learning outcomes.

Across primary and post-primary level, there needs to be clear communication of the rationales and motivations for integrated STEM education. At post-primary level, integrated STEM options in first and transition years should be expanded. At post-primary level, senior cycle reform needs to be implemented, along with reform of the leaving certificate. The requirements at junior cycle could also be revisited. Physical resourcing must also be considered. Spaces for STEM in early years and primary settings are currently limited, while post-primary students require access to well-resourced laboratories so they can develop appropriate laboratory skills and engage in the types of activities outlined in the subject specifications.

Given the evidence of problems with teacher confidence related to STEM, as mentioned by some of my colleagues, engagement by universities and schools with stakeholders in the community and industry offer innovative ways of building teacher confidence. One such initiative is the STEM teacher internship, STINT, programme. It began in DCU and has expanded nationally. It provides pre-service teachers with the opportunity to avail of internships in industry. They can gain hands-on experience of a wide range of careers and opportunities available within STEM industries. The evidence from the programme shows that participation in STEM roles during pre-service teacher education can significantly impact teachers' understanding of STEM and STEM careers and influence their future classroom practice.

Student participation in STEM events such as Scifest, the BT Young Scientist and Technology Exhibition, etc., should continue to be encouraged because that develops confidence and interest in STEM. I again thank the committee for the opportunity to contribute.

Professor Áine Hyland: I thank the Cathaoirleach and members of the committee for inviting me to take part in this round-table discussion on the future of STEM in Irish education. I also thank the committee for the report of May 2022 on senior cycle reform, Learning for Life, which provided an excellent overview of the issues and concerns raised by teachers. That report has been widely welcomed.

I support the statement made by Humphrey Jones on behalf of the Irish Science Teachers Association at the session of the committee on 21 March, where he set out the concerns of science teachers about the new junior cycle science and leaving certificate agricultural science syllabi. He expressed the hope that the NCCA will listen to the voices of science teachers before it finalises the proposed new leaving certificate physics, chemistry and biology syllabi.

I am particularly pleased to have the opportunity today to draw the attention of the committee to a relevant and exciting new initiative, namely, the national children's science centre, also called the museum of possibilities. I am the chair of the exhibits advisory committee. This project has been in gestation for more than 20 years and is finally about to come to fruition. This new science centre will support and complement the Department of Education's primary and post-primary science curriculum and will provide facilities for students and teachers at all levels, from early childhood to leaving certificate level, to engage in an interactive and non-formal way with the world of science and its possibilities.

Until now, Ireland has been the only country in the OECD and EU not to have a national children's science centre. The need for such a science centre was identified in 2002 by the then Irish Council for Science, Technology and Innovation and was reiterated in the report of the review group on STEM education in the Irish school system in 2016.

The national children's science centre aligns with the Government's STEM Education Policy Statement 2017-2026, which emphasises the need to nurture "curiosity, inquiry, problem-solving, ethical behaviour, confidence, and persistence, along with the excitement of collaborative innovation". It will also contribute to the implementation of the UN's sustainable development goals, SDGs, and it will be a resource for students and teachers in the interpretation and expansion of the Department of Education's STEM curriculum at preschool, primary and post-primary levels.

The OPW has drawn up a stunning plan to restore and refurbish the north wing of the National Concert Hall on Earlsfort Terrace, and planning permission has recently been granted by Dublin City Council for the national children's science centre in this location. The centre,

totalling almost 10,000 sq. m, will house three floors of immersive, interactive exhibits, a laboratory and a purpose-built, state-of-the-art planetarium. It will have dedicated activity spaces for very young children and it will cater for primary and second-level pupils, as well as for the curious of all ages. It will have a policy of equality of access and it will have a particular focus on ensuring ready access for people of different abilities and disabilities and for those from less advantaged backgrounds.

The board of the national children's science centre, which is a voluntary group of people committed to education in STEM and the arts, is chaired by Michael Collins and includes the eminent scientists, Professor Luke O'Neill of Trinity College Dublin and Professor Brian Ó Gallachóir of University College Cork, as well as Ali Hewson, Jonathan Westrup, Mindy O'Brien, Cathy Moore and Frank Doonan. The centre will adjoin the Iveagh Gardens and will re-establish the original connection between the building and the gardens, restoring it to the way it was 160 years ago when the building was used for the International Exhibition of Arts and Manufactures. The building will not, however, encroach on the gardens themselves in any way.

In the early 2000s, the Government agreed to co-operate with the board of the national children's science centre in delivering such a centre by providing a venue and a suitable home for the centre. The original location was to have been in Heuston Gate, but that plan fell through during the recession. The current location in Earlsfort Terrace is in a wing of the National Concert Hall that has been unoccupied since the medical faculty of UCD moved to Belfield in 2007. An agreement was reached in 2016 that the OPW would conserve and renovate the building and that the board of the national children's science centre would fund the exhibits and set up a structure to manage and run the centre. We are very excited about this development, which we know will greatly increase the level of interest in STEM and STEAM education for all.

We would very much welcome the support and endorsement of the joint Oireachtas committee for the proposed new centre on Earlsfort Terrace.

An Cathaoirleach: On the last day Professor Hyland was here she shot from the hip and I have no doubt she will be doing the same today.

The first member up is Deputy Mairéad Farrell, who is substituting for Deputy Rose Conway-Walsh. She will be followed by Deputy Jim O'Callaghan.

Deputy Mairéad Farrell: Gabhaim buíochas leis na finnétithe as teacht os comhair an choiste. In particular, I thank the Cathaoirleach for welcoming us in. This is my first education committee meeting. I will assume it is always only women who come before the committee and I will be expecting nothing less. I am aware that I have approximately six minutes, so I will try to keep to the point. This has been very interesting. As this is my first time at the education committee, if I do ask obvious questions, I hope those present will pay no heed. It has been fascinating to read these opening statements.

I have two questions for Dr. Marcus-Quinn. What she said in relation to the requirement for students to have their own devices was very interesting. She said that when the Department of Education embedded the use of technology into the junior cycle, the provision of adequate access to technology was not sufficiently considered in the budgets that were available to schools. She mentioned the Barnardos' survey. Of course, we are all aware of the outcomes of that particular survey. I am interested in hearing whether much analysis has been done on the socioeconomic backgrounds of people who are going further into STEM. There has been

a gender-based analysis, but has there been an analysis of socioeconomic backgrounds, given the mention of cost?

Dr. Marcus-Quinn spoke about the tendering process for the purchase of devices. I focused on procurement previously. I understand that each school is considered its own contracting authority. Under EU public procurement regulations, we would be permitted a vehicle whereby schools could come together and contract. Dr. Marcus-Quinn mentioned that there was no tendering process that would allow for cost savings. What are her views on this matter?

Dr. Marcus-Quinn stated that Australia's 2008 Digital Education Revolution programme was worth considering. Will she provide further details on that?

Dr. Ann Marcus-Quinn: I will take the Deputy's second question first, if I may. Schools get some grants for the provision of digital devices at post-primary level. There has been a welcome increase in that funding in recent years, particularly since the pandemic, but how that process is treated differs from the process where devices are purchased by parents. When public money is involved, a formal tender process must be gone through, but when the devices are funded by parents, there is nothing in place. A school can decide to roll out whatever plan or policy it wants. There is nothing to stop an individual principal or school from dictating that each incoming student must buy a top-of-the-range Apple or Microsoft product even when the student's requirements may not necessitate such a device and something more basic would fulfil the same role, for example, a Chromebook or, in early years education, a Raspberry Pi.

We see a different attitude towards what is expected of parents' spending between primary and secondary levels. In primary school, parents typically do not have to furnish their child with a device and the purse is more wisely spent. For example, there is greater use of Chromebooks and such and there is a shared trolley system. Children may be using tablets, but everything is much more organised. How schools operate at primary level is different because the teacher has a base classroom and things can be stored within it.

We need to examine how schools are making decisions on what devices they are buying, as there is inequality. Where schools have a bring-your-own-device policy, there is a significant mix in the classroom. In the consultation on a digital strategy, approximately half of students reported that they had a bring your own device system in place, with a mix of laptops, tablets and smartphones being used in the classroom. There is little equity, though. In some cases, one child uses a damaged screen in battery stamina mode to complete the spot Kahoot! - Kahoot! is a popular programme – assigned to the class while another in the class uses a top-of-the-range Microsoft Surface or the like with a large screen. There is no comparison between those two teaching and learning experiences in the same classroom, yet the children will face the same assessment. There is an issue with a minimum standard of quality and what people are expected to purchase.

The Deputy's first question was on how many students went on to study STEM subjects. I am sorry, but that is outside my area and I am unsure as to how many go on to study STEM.

Deputy Mairéad Farrell: That is fair.

Dr. Ann Marcus-Quinn: If a student in post-primary has a poor experience of STEM subjects and that poor experience is largely down to the device he or she has had to use, it will impact on the student's choice of course of study at third level. We need to examine the issue of equity of device experience at second level.

Deputy Mairéad Farrell: I have a question for Dr. Connolly. Her mention of the difficulties in, and barriers to, getting people to become computer science teachers was interesting. It was fascinating that, of the 140 people teaching, the vast majority do not have accreditation in the subject. According to another opening statement, 25% of post-primary mathematics teachers in Ireland do not have a mathematics qualification. Dr. Connolly observed there is a barrier which experienced graduates face with the two years which stops them being able to qualify. Looking at how we can remove those barriers, I was thinking about people who might have done a PhD and have teaching experience as a result of that. Would she consider someone who has a PhD and that experience having to do a two-year course a barrier? Could someone with that kind of a background do a shorter course to become a qualified teacher?

Dr. Cornelia Connolly: There is a shortage of teachers in computer science education globally. We have to think creatively around how to upskill teachers. There is a very successful professional diploma for mathematics teaching, which is ongoing over recent years through University of Limerick and University of Galway. Offering a similar programme to those teachers in our system to upskill and become qualified computer science teachers would be an easy win. We need, however, to incentivise people into computer science education and encourage more into the education space. The internship programmes for student teachers is an obvious choice, as is expanding and showing the range of skills possible within computer science education.

We must also be conscious of the learning pathway. It is very difficult for students to select computer science or a STEM teacher education programme coming from their leaving certificate if they have not studied the subject in secondary school themselves. There are many variables or little challenges we have to overcome.

Deputy Jim O’Callaghan: I thank the panel for coming before the committee and for their very helpful opening statements. I will start with Dr. Connolly on computer science. I have to confess my own ignorance here. I had assumed when we are talking about STEM that a student in secondary school who had done other science, technology, engineering or maths would come out with an understanding of computers and their use. Am I wrong? Is it necessary for them to do computer science to get the level of detail which Dr. Connolly specified?

Dr. Cornelia Connolly: As I tried to outline in my opening statement, an easy example would be where a Deputy or Senator is trying to circulate his or her election manifesto via email. That is digital skills which is available to all students in our school system. However, the understanding of an algorithm to write a piece of code or to use code to analyse voting patterns or monitor constituent concerns are the skills involved in computer science and those skills are not available to every student in the country. We have conducted research that highlights the issue of equity with the rural-urban divide. That touches a little on Deputy Farrell’s question earlier. The socioeconomic divide is becoming transparent as well as gender issues around STEM education and computer science education. These are challenges that are emerging in the small number of schools that are currently offering computer science education.

Returning to Deputy O’Callaghan’s questions, it is around computational thinking, coding skills, the understanding of algorithms and abstract thinking. They are the skills that are purely related to computer science as opposed to digital.

Deputy Jim O’Callaghan: When can that start? Dr. Connolly mentioned there is coding that can be done at primary school. Is that something parents do voluntarily? It is not something the schools provide. Is that correct?

Dr. Cornelia Connolly: My colleagues have already touched on it but, with Aistear and the new primary curriculum framework, we have an opportunity with the competency framework to ensure coding is introduced at primary level. Basic coding like Scratch, drag-and-drop or the use of Raspberry Pi could be introduced in the primary framework. As a result, students would have comprehension of coding going into post-primary.

Deputy Jim O'Callaghan: Dr. Marcus-Quinn mentioned the scenarios that operate in schools when it comes the availability of devices. Is there a role for the vibrant tech sector we have in Ireland to assist the education system in schools in terms of the provision of devices? I am conscious that during the pandemic Google was helpful in my constituency in terms of providing devices to local schools so kids would have access to them.

Dr. Ann Marcus-Quinn: Absolutely. It should not be about the placement of a brand within a school. It is worrying when we see schools announcing themselves as a Google school, an Apple school or whatever. We never heard teachers announcing themselves as a Folens or Macmillan teacher. I do not see how there is any real difference there.

There is certainly scope to have a better roll-out of similar appropriate devices across all schools. If we could agree on a minimum standard, what the brand is should not matter as long as all students had a uniform device in front of them so they could complete their classroom activities and whatever assessed work they needed to do.

Deputy Jim O'Callaghan: Is that something that should be co-ordinated centrally by Government?

Dr. Ann Marcus-Quinn: Absolutely. For third level, we have HEAnet and an online purchasing option. If faculty or people working in the third-level sector want to buy devices they can go through that. We should roll out something similar for post-primary.

Deputy Jim O'Callaghan: Professor McCormack was correct to emphasise the significant performance by women who are doing engineering courses. One of the things we have considered over the past number of weeks is how we can encourage more women to take up STEM subjects. We were given a statistic that approximately 40% of the PhDs being done in STEM subjects are being completed by women. Does that operate at the higher levels of the profession? In the professorial constituents of the engineering department in Trinity or other STEM subjects, are women equally represented at those levels?

Professor Sarah McCormack: I thank the Deputy for the question. I appreciate that he is aware of the statistics, which is good. Women tend to have higher entry going into engineering. Students really want to do the subject and are really committed. In terms of how that goes through the pipeline, we have what is known as a leaky pipeline along that progression. We see a lot of drop-off, in particular from entry-level lecturing positions and trying to maintain that. Where a student has completed a PhD and is moving into a research position is a critical moment when there is no real career path. We do not have a huge number of academic jobs available to those people, something which affects both genders. From a personal perspective, I find as women meet someone and have families they find it very difficult to continue on that research track and maintain their CV in order to become an academic in the future.

Deputy Jim O'Callaghan: The report mentions that it is important to instill an interest at second level. What can the committee recommend to ensure we try to get greater participation among women in studying STEM subjects in secondary schools?

Professor Sarah McCormack: It involves trying to give women the opportunity to study STEM subjects. Some schools do not provide the opportunity to study higher level maths or physics. Students sometimes have to attend outside classes for such subjects. It is about giving equal opportunity for those subjects. This should not be seen as something different for women to do. It is unexpected for a woman to want to do that. It is about trying to break down those barriers.

Deputy Jim O’Callaghan: I have run out of time. I cannot ask any more questions.

Senator Eileen Flynn: I have a general question on leaving certificate reform. How do people think that should look in terms of subjects? If we reformed the leaving certificate, what would science and maths look like? When we talk about STEM, in my opinion, we are talking about gender equality and equality in general. I see very little opportunity for people from minority groups. Here I am again at the Joint Committee on Education, Further and Higher Education, Research, Innovation and Science talking about the others. Where is Traveller participation or that of migrants, refugees and people with disabilities? What does it look like for those students to have equal access and be supported? It is one thing for a member of the Traveller community to go on and complete the leaving certificate or even complete primary school, and it is different again going to secondary school. What supports are available for young members of the Traveller community to participate in STEM? In 2023, how can we break down the cultural barriers to women participating in these courses or subjects? What do we need to do as a State to encourage more participation by minority groups and, of course, people with disabilities, in science subjects? I spoke previously in this committee about the digital divide and the gap between poor and rich children being able to access technology. One story that struck me was about a Traveller woman who had to use one tablet during the pandemic for five of her children and that of another Traveller woman who had to use her own phone for three of her children. Although we like to think we are closing the gap in the digital divide, I do not think we are on the ground. If we want children to participate in science subjects and want them to take part, how do we support them? One shoe does not fit every child.

I agree with the science centre for young children and the participation of youngsters because you are never too young, but that starts at home and the supports parents need to encourage their children to take on maths, science, computer science, whatever it may be, because it starts from a young age and develops on. Will the witnesses comment on those remarks? I am very interested in leaving certificate reform and what it would look like.

Professor Áine Hyland: On leaving certificate reform, it is important that there are many examples and a lot of detail. It was already raised in the Learning for Life report by the Oireachtas committee that broad guidelines are not enough and there needs to be more information, materials and resources, which others have mentioned. Regarding the national children’s science centre, we are focused on not just primary and second-level children but children from preschool as well. There will be a special space set aside for those children. We also have a strong commitment to inclusion and to ensuring that not only will it be accessible to all, whatever their background, be it Traveller, refugee, and the whole equality, diversity and inclusion concept, but also to children who are neurodiverse or children with disabilities. The exhibits will be designed using universal design for learning, which will ensure inclusion insofar as possible. We are very conscious of all of these issues. I thank the Senator for asking the question.

Dr. Cornelia Connolly: It is a great question. Regarding leaving certificate reform, there is an opportunity to examine inquiry-based learning and project-based learning more and expanding those also in terms of assessment. For example, in the design of the leaving certificate

computer science curriculum, one was restricted because the exam had to count for 30%, even though it is very much a hands-on subject. There are restrictions that require more flexibility. The digital divide is a problem but there is also a digital use divide. It is not sufficient to give the device to schools, communities or families. We must ensure they can use the device appropriately. I am conscious of the digital divide but the digital use divide is equally important.

Many universities have ongoing outreach projects. For example, the University of Galway is involved with Trinity and the University of Limerick in the CodePlus project, which targets DEIS schools, transition year students and young girls to introduce them to coding in communities where they would not usually get that opportunity. The CodePlus project has been ongoing in recent years. We are now examining it from a longitudinal point of view to see how it has affected girls' choices in subject and degree programme at university. Such projects are important. It is difficult for teachers to do everything within school hours, so opportunities like that are worthwhile.

Dr. Margaret Leahy: The question of addressing most of the issues of inequity and inclusion to which the Senator referred goes back to teacher education. I was about to say that the digital divide was about more than the provision of devices to children and getting devices into their hands, but about how they used them and evaluated and developed critical thinking skills around the use of that technology. In addition to outreach, that has to involve our schools. We need teachers who know how to use these digital devices or digital learning as an integral part of what they are doing in schools. It has to start with the teacher. I am sure Professor Venkat would say that STEM and digital knowledge has to begin at the very start in early childhood and primary school. Get children young and develop those skills and thinking so that we can progress them across the continuum.

I will make a comment about leaving certificate reform and the question of whether the tail is wagging the dog. Is it assessment or the programme? We do policy and curriculum well, but the traditional leaving certificate still involves rote learning and we know that teachers teach to the test. If we want to embed our curriculum reform, we need to consider what we are assessing at senior cycle in order to promote the kinds of thinker we are hoping to develop.

Senator Eileen Flynn: Did Professor Venkat wish to come in on that?

Professor Hamsa Venkat: It was simply to echo the point that a great deal of evidence suggests that we have to intervene early. Evidence in Ireland and internationally is showing that, by the end of primary school, attitudes have formed. It is difficult to do things in secondary education. It is almost a case of using sticking plasters to try to address a much larger problem. We cannot get students in at that point if they have formed negative attitudes.

An Cathaoirleach: Next are Deputy Clarke and Senator Pauline O'Reilly.

Deputy Sorca Clarke: I thank the witnesses for attending. This is one of the rare occasions when I agree with everything that people on the other side of the room have said. I make no apologies for that. I love STEM. The witnesses can ask Deputy Farrell, who is sitting beside me. I am an unrepentant geek. However, it takes more than being a geek to have an interest in STEM subjects.

I was delighted to hear Dr. Connolly trying to break down the mystery of computer science. In some cases, it is almost seen as the third secret of Fatima, but that is not what it is. It is a stand-alone subject and needs to be valued as such.

Dr. Connolly spoke about equity and equality in access to certain devices through schools and the various options available. The majority of children are educated in schools, but not all are. What impact does the lack of standardised access to devices have on children who are home-schooled, which depends on the ability of the parent? Are additional resources available through the Department of Education for the parent or guardian?

I do not have much time, and I wanted to give more time to each of the witnesses to speak, but I will focus on issues raised by Dr. Leahy and Professor Venkat. Recently, the second STEM education implementation plan was published. It covered the years 2017 to 2026. To my mind, that is an extremely long period. Does Professor Venkat have an opinion on that? If the length of time in which science is taught in school was to be extended, is the curriculum fit for purpose to integrate the discipline and integrated aspects of STEM and to give confidence to students from a young age to leaving certificate level?

Professor Hamsa Venkat: I will begin with the question relating to the length of time dedicated to science. In the new primary curriculum, there is a monthly specification that goes with science, technology and engineering. The danger with that, as we have seen even before the introduction of this curriculum, is that science time is already being whittled down. Teachers who are not very confident about their own knowledge of science are being told they will have to teach science, technology and engineering in the same time allocation. We think science is a very important part of STEM, but if we want to protect it, we will have to give enough time to it.

Our suggestion would be to give science a weekly time allocation and, moreover, to give at least termly access to an extensive, expansive STEM project, of various sorts, that children at all grades, from early years into primary schooling, would get every year. They are going to complete a range of projects. We would strongly say teacher education needs to come on board. All of us around this table, I would guess, got into STEM on the back of traditional disciplinary learning. We did not have much access to integrated learning. Integrated learning is new for many teachers, especially in the field of STEM, so teacher education should work in parallel.

I agree that extending the programme to 2026 sounds like a long timeline, but in educational innovation anywhere in the world, it is probably not long. What we would not want is to do is get to 2026 and find we are not much further on than we were in 2017, so thinking carefully about what we put in place systemically in teacher education is a major part of this. Having said that, so too is the provision of technology and digital resources that we need in schools. I would suggest that to do that carefully and systemically, we will do well to think in the medium term rather than in a shorter timeline.

Deputy Sorca Clarke: Professor McCormack spoke about the societal and cultural bias and the conscious and unconscious elements of that. I sat on the Joint Committee on Gender Equality, which reported just before Christmas. This was an area of which the citizens' assembly recommended further examination. One issue that came through strongly at the Joint Committee on Gender Equality related to the fact we pigeonhole students from a young age. By the time they get to fifth and sixth class and are considering which secondary schools to go on to, there is an element of geographic lottery to the range of subjects that are available and to whether the student will be able to study STEM subjects in secondary school. In the experience of Professor McCormack and all our guests, how important is it to increase access to subjects among children from fifth and sixth class?

Professor Sarah McCormack: That is one of the key things we need to do to. We need to ensure all students have equal access and opportunity in regard not only to STEM but to other

subjects as well. Transition year students come to us on workshops and we often hear they have to study physics outside of their own school. It is quite difficult to come from a school where there is not enough interest, but it is about them having that understanding of what physics is and seeing that it is no more difficult than other subjects. Sometimes there is an idea, even going back to primary school, that certain students are not good at maths or science and they might pick biology because they think physics is too difficult. These are misconceptions some people hold but they might not know enough about physics to say whether it is too difficult. Breaking down those biases regarding the subjects is important.

Likewise, in the case of computer science, the mention of algorithms is enough to turn some students off the subject. Explaining to them what an algorithm is and how they are used in everything we do today is important, and it is about getting in early to explain what these subjects are. We often see the unconscious bias stuff whereby people are waiting on the firefighter to come in. They are expecting men and it turns out to be women, and the little kids are so surprised because they have never seen a woman firefighter before. With these types of things, it is very important that we try to get in as early as possible so they understand the opportunity is there for everyone and it is not just men's career choices and women's career choices.

Deputy Sorca Clarke: The concern I have regarding a student having the option to do a subject outside of school is that as there is also a student who does not have that option, we are further embedding inequality. Parents have to have access to the resources, to the transport and to the time. There is an additional pressure put on those who are least in a position to be able to meet those additional asks of doing a programme outside the set school hours.

Dr. Cornelia Connolly: In the 1980s, the Department had a programme to provide physics to female-only schools, so the sharing of teachers is an option, for example, in physics or computer science, or it is an idea.

Senator Pauline O'Reilly: I thank the witnesses for their contributions. I have read all of their opening statements. Education is about two things, namely, the personal benefit for one's own well-being and the social good. I would like to address both. On the first, the whole question of equity of access has been very well covered in many of the opening statements, in particular by Dr. Marcus-Quinn, Dr. Leahy and Dr. Venkat.

I sat on the Joint Committee on Gender Equality. I am very conscious of the gender inequalities when it comes to STEM. I am also very concerned about the level of teaching, or if that is the wrong word, then the level that courses are at for junior certificate. Professor Hyland said she agreed with what had come before us previously from the science teachers organisation. Dr. Leahy also mentioned this idea that the exam-based system was masking some of the failures in the system and that inquiry-based learning was dropping out. When it comes to science, innovation and computer technology, it has to be about inquiry because many of the questions and many of the facts are unknown at this point and it is really about developing a mindset. I have heard that Ireland has a very low level of innovation in the SME sector, especially when compared with others in Europe. I wonder if that lack of innovation is reflective of the education system, particularly around the sciences. I would like to get the witnesses' views on that matter.

I have been the chair of the Home Education Network. I also served as chair of a school board of management. Somewhere in the middle lies the answer. The sense of inquiry and the love of learning that we get with home schooling matched with the mentors, who would have a very full understanding of what sciences are. That is the sweet spot and I do not feel the educa-

tion system has hit that, particularly when it comes to STEM. Perhaps that is why girls are not engaging with it to the extent that they should be.

Professor Áine Hyland: I agree. There is a mismatch in a way between current developments, such as the changes in the junior cycle and leaving certificate, and the examination and assessment, the State Examinations Commission and the NCCA, which has been pointed out before. There are also the very skeletal programmes, syllabi or specifications, as they are called, that are coming out now for the proposed new leaving certificate subjects. I do not think they give enough information to teachers and they do not go into sufficient depth. There is a real risk that standards will begin to fall. The committee has heard this from the Irish Science Teachers Association.

Regarding love of the subject, I have six grandsons and have brought them to science museums and science centres in different countries. It really sets their interest and enthusiasm on fire when they are in a non-school environment, as well as being in school. The additional impetus, interest and enthusiasm that one can see emerging when they are in a children's science centre where they are completely free to interact with different exhibits, and where they have support from mentors of different ages and stages within the science centres, really sets them on fire to some extent. It is great. This is an investment that Ireland can certainly afford and from which it will get a return. The Minister who supports this and goes with it will be the Donogh O'Malley of the 21st century. We are committed. The STEM education implementation plan contains a total commitment by Government regarding STEM education. We have heard about the areas in schooling where some extra investment would be helpful. This relatively small investment will be matched by private investment. This is a social enterprise. It would really give a boost to STEM education.

Senator Pauline O'Reilly: It is interesting and exciting. Are there any other comments?

Dr. Margaret Leahy: I will go back to early childhood and primary education again and think of the love of a subject that teachers can engender in their pupils. Time and time again, we hear that people did a subject because they had a wonderful teacher. Early childhood and primary teaching are predominantly female professions. We have been talking about the lack of female participation in science at second level or third level. A number of people who are coming into these professions do not have a solid foundation or even interest. Those of us at pre-service level need to take these students, develop their understanding and knowledge and give them as much experience as we possibly can in order that they, in turn, can engender a love of STEM, embed STEM and do STEM in practice.

Dr. Margaret Leahy: I thank Dr. Leahy. I am out of time. I appreciated Professor Venkat's contribution too. What we will take away from today is the emphasis on primary education being so important to STEM. I thank the witnesses for their contributions.

Senator Rónán Mullen: Cuirim fáilte roimh na finnétithe agus gabhaim buíochas leo as a saineolas a roinnt linn. I am grateful to the witnesses for their impressive presentations. I do not know whether to be happy or sad about the fact that Raspberry Pi seems to have moved out of the realm of domestic science and into that of computer science. It appears that virtual reality can only ever be virtual. On that subject, are the witnesses worried that something is lost through the use of the STEM acronym? Are they advocating a STEM-plus way of looking at the world? I see their figures on the number of students getting to do computer science in school are low.

Dr. Cornelia Connolly: It is something that one should be aware of, namely, that computer science is not included in the STEM acronym. I could stand to be corrected but, for example, there is a Computer Science Teachers Association. I am not sure if it was invited to make a presentation to the committee. There are issues like that. At times, we miss out computing when we talk about STEM. It is important to note that the fundamental skills of computational thinking, as we have all reiterated, need to be introduced at an early stage, at day or preschool and in primary, then in post-primary. We have a leaving certificate subject at the end of our education system, but we have not facilitated the learning pathway.

Senator Rónán Mullen: Dr. Connolly said 15% of schools offer it.

Dr. Cornelia Connolly: Yes.

Senator Rónán Mullen: Dr. Connolly said 22% of girls do it. Is she saying 22% of girls overall in the country or that 22% of those who do it are female?

Dr. Cornelia Connolly: I cannot recall.

Senator Rónán Mullen: That would be a higher percentage of girls than schools. I was wondering about that.

Dr. Cornelia Connolly: Some 22% of the leaving certificate numbers for that year were female. We obviously have a gender issue.

Senator Rónán Mullen: Within the schools, a much higher percentage of girls are doing it than guys.

Dr. Cornelia Connolly: It is also worth noting the urban-rural divide for availability to do the subject. That also relates to the socioeconomic locations of schools.

Senator Rónán Mullen: I will not be able to get to everybody. All of the witnesses said interesting things, but time will prevent me from getting around to all of them. Professor McCormack spoke about women's attainment. It is clear and impressive that more women come out on top in engineering at the highest level. The completion rate is 90% compared with 67% for boys. Another way of looking at that is to ask whether we should be concerned about men's attainment here. It is right and proper to talk about getting more women interested in STEM, but we have to have equal concern. What is going on here? Is it that there is such a bias in favour of boys doing these subjects that more go for it who are not really as suited for it and the small cohort of women who go for it are those who are really good? Is that what the figures are telling us?

Professor Sarah McCormack: Yes.

Senator Rónán Mullen: Does Professor McCormack believe that if she controlled for everything, there would be any likelihood of a greater level of aptitude among boys in general? Does she think, everything being equal, it would be 50-50 for aptitude and ability in these areas?

Professor Sarah McCormack: Is Senator Mullen asking if women are better than men?

Senator Rónán Mullen: I am asking if there is a difference between boys and girls when it comes to these subjects.

Professor Sarah McCormack: Not necessarily. I do not think so. For these numbers, the

women who choose to do engineering have come in with high grades. They are committed to the subject. We are trying to increase the number of women coming in. The men who are coming in do not have points as high as the women. That is where the difference in grades is coming out at the end. We are trying to encourage more women in. They do not have to have the top grades to come in but those women are thinking in some ways that they are not able to do engineering because they do not have very high grades. We want to show the men coming in with certain grades and that if women came in with the same grades, they would have the same aptitude to do these subjects.

Senator Rónán Mullen: I am excited by the national children's science centre. We all wish it a fair wind. I have a couple of questions on that. Is there any overlap between it and what has been going on in the science centre or science museum in Trinity College Dublin?

Professor Áine Hyland: The Science Gallery.

Senator Rónán Mullen: The Science Gallery, I meant to say, which housed, in a public interest, a collection of material relating to science.

Professor Áine Hyland: The focus of the Science Gallery was somewhat different. It was for older teenagers and adults. It was also entirely topic-based. It was a good centre. We do not know whether it will be reopening. We hope it will. There will be much more focus on preschool, primary and post-primary pupils if it does.

Senator Rónán Mullen: So Professor Hyland would not see a significant degree of overlap?

Professor Áine Hyland: No. It will have permanent exhibits, and we plan for it to provide support for teachers, with much extracurricular work. It is tied into the curriculum in one way and would expand the curriculum for schools that do not have the resources and may not have suitable laboratories. It would be an opportunity for children or young people who are being home-schooled. Various children's science centres internationally provide much online support too. It is ambitious. It does not pretend not to be ambitious. The amount of money that is being raised and was originally pledged to the museum when it was to have been located at Heuston Gate was €10 million. We are talking big money. It is a large centre, at 10,000 sq. m. We have the support of the OPW and its architects. The OPW is providing the building. In due course, we hope that the Government will come in behind it.

Senator Rónán Mullen: I was glad that Professor Hyland referred to the Government's STEM education policy statement about nurturing ethical behaviour, among other things. I always think of Churchill's dictum that scientists should be on tap but not on top. How will the national children's science centre deliver on those ethical ideas? It may be a question for all the witnesses because we did not touch on the issues of ethics. We only have to think of the history of the 20th century and the behaviour of German doctors in concentration camps, for example. The reality is that there is knowledge that we use today and benefit from that was obtained unethically. There are many areas of contested science such as embryo research, all sorts of issues that are very highly charged, both politically and ethically.

An Cathaoirleach: Briefly, please.

Senator Rónán Mullen: How might the centre deliver on this and how do any of our guests think schools should approach it? Is it approached through religion class or ethics classes, or should it be handled by teachers of science? Thoughts, please, because we did not discuss the

ethics side very much.

Professor Áine Hyland: This is a very important point considering where we are with science and technology. ChatGPT is an indication of the challenges that are before us.

Senator Rónán Mullen: Artificial intelligence.

Professor Áine Hyland: Yes, and, as the Senator says, all the DNA research. It is multifaceted. All subjects need to address it. It would not be sufficient to have issues of ethics addressed just in religion class, for example. It must be integrated more. When the NCCA comes up with a new syllabus it really needs to provide examples. Somebody else has mentioned this. Teachers should get some indication of how they are to approach these issues. If the NCCA does not do it the textbook writers will, and they may forget to put in anything about the ethical issues. I think ethical issues are very important.

An Cathaoirleach: Senator Maria Byrne is next. She will have three minutes rather than six because she is not a committee member.

Senator Maria Byrne: I thank the presenters. I heard some of the statements from my office although I had to go to something else earlier. I would like to address my questions to Dr. Marcus-Quinn. Why does digital policy differ between schools in respect of devices? I am involved with a school in which iPhones and iPads are prominent. They are used for everything, including receiving books and accessing whiteboards in order that pupils can get information and access extracurricular support and advice. It differs between teachers and students. Would they prefer to have smartphones for classwork? Dr. Marcus-Quinn spoke about a centralised approach to the procurement process of digital devices. Could she explain how that might work?

Dr. Ann Marcus-Quinn: I have a quote from our Minister for Education. I figured I would probably be using it at some point this morning. Public schools in Ireland are autonomous. In that context, the Minister has stated, “decisions regarding the deployment and use of digital technologies are a matter for the management of schools as they are best placed to determine this according to their own situation and requirements”. This stance would have been fitting before technology was embedded across the six key skills of junior cycle. Now that all students are required to use technology to some degree in their classroom activities and when participating in State assessments, it is time to look at the level of autonomy the schools have and to form a more equitable experience for all students who have to use technology across all subjects. I also think that the parent voice is not strong enough. The digital policy is such an important document but very few schools update it as frequently as they should. Many schools have a digital policy that has been *in situ* since before Covid. We know how much the world has changed in terms of schooling and digital devices since then. When it came to choosing a secondary school for my eldest daughter, the digital policy was certainly foremost in our minds. I do not think schools recognise how important that document is in influencing parental decisions on where they are going to send their children. As the ones paying for the device in many cases, they should have more of a say on what is used in schools.

Senator Maria Byrne: It varies from school to school here. Are there any examples from Europe or elsewhere that we could look at in order to form policy?

Dr. Ann Marcus-Quinn: It is interesting to see how rapidly the technology is moving. In the review of the use of tablet devices in schools three years ago, we recommended trolleys as

a solution. This would maybe not be the best solution for today, because there is such pressure in schools. All classes need access to devices. It is great that teachers can spontaneously plan to use technology within a classroom setting. There is much more pressure on schools in terms of the technology they are using. There are two really good examples, though. If we look as far back as 1997, there was the Estonia Tiger Leap project, which set out to provide devices to teachers and students. More recently, there was a digital revolution in Australia whereby the authorities there looked at putting devices into the hands of teachers and students. Again, it was a more equitable experience. I would like to emphasise, because technology is moving so rapidly, what would be an ideal solution this year may not necessarily be the best solution next year or as we move forward. It is something to monitor continually.

An Cathaoirleach: I apologise that the witnesses for the second session will be coming into the committee room. People have been waiting outside for a considerable length of time. This committee session must be over by 1.30 p.m. as well.

I thank the witnesses before us for attending. This discussion has been very productive and will be of great importance in terms of the committee's input on STEM in the report we will be compiling on the basis of our hearings in recent weeks. I appreciate the witnesses sharing their expert knowledge and research with the committee. I thank them very much. We will suspend proceedings for a moment.

Sitting suspended at 12.41 p.m. and resumed at 12.45 p.m.

An Cathaoirleach: We will resume. On behalf of the committee I welcome our guests. From WorldSkills Ireland, Mr. Jamie Bermingham is a plumber and World Skills Ireland plumbing and heating champion 2022; and Mr. Martin Scattergood is a sheet metal work apprentice. In the Public Gallery we have Mr. Donal Keys, Mr. Fergus McGuinness and Mr. Eamonn McDonnell, also from WorldSkills Ireland. The witnesses are here today to discuss the future of science, technology, engineering and maths in Irish education. The format of the meeting is that I will invite Mr. Bermingham and Mr. Scattergood to make brief opening statements. That will be followed by questions from members about their experience of what they are doing at the moment.

I remind the witnesses that they should not criticise nor make charges against any person or entity by name or in any way as to make him, her or it identifiable or otherwise engage in speech that might be regarded as damaging to the good name of the person or entity. Therefore if their statements are potentially defamatory in relation to a person or entity, they will be directed by the Chair to discontinue their remarks and it is imperative that they do so.

I invite Mr. Bermingham to make his opening statement.

Mr. Jamie Bermingham: I am a newly qualified plumber and WorldSkills Ireland plumbing and heating champion 2022. I would like to begin by saying that my apprenticeship has been a life-changing experience for me. Not only has it helped me to acquire the necessary skills and knowledge to excel in my chosen field, but it has also opened doors to numerous opportunities and experiences that I never thought were possible. Growing up, I spent a lot of time around cars in the family garage and had a deep passion for them. I loved working with my hands, solving issues, learning how things work and operate. I thrived with the satisfaction of fixing something that was not working or was broken. I knew for sure that pursuing an apprenticeship as a motor mechanic as soon as I finished secondary school was the path I was going to take, and that third level was not what I wanted. Unfortunately, when I finished school, it did

not work out for me, and I could not seem to find anyone or anywhere to get an apprenticeship in the motor trade. I was very upset and disheartened, but little did I know what was in store for me down the line.

My parents understood my passion was cars, but they advised me to maybe try something different, like an electrical or plumbing apprenticeship. I started out in the plumbing trade working for a relative, mainly in the domestic end of plumbing. However, it was not until I started my apprenticeship that I realised the true potential of the industry and the opportunities it can provide. Later in my apprenticeship I joined the Jones Engineering Group to gain more experience, and it has been a rollercoaster of experiences since, starting out on the construction side then moving to the maintenance division and making a name for myself through the interest I showed and the quality of my work, which led me to where I am today.

I am proud to say that I succeeded well in my apprenticeship, achieving distinctions in all my exams and qualifying with an overall distinction. This was a moment of great pride for me, as I knew that I had worked hard and dedicated myself to the apprenticeship programme. My achievements did not stop there. I was privileged enough to compete in the WorldSkills Ireland heating and plumbing competition in the RDS last year, where I was able to showcase my skills and compete against other talented apprentice plumbers from around the country. This was an incredible experience, and after three intense days of competing, I was thrilled to win the event. It was a humbling experience to be recognised for my skills on a national platform, and it was an honour to represent myself and my company in such a prestigious event. In addition to this, I had the opportunity to take part in the Plumbing Champions event in Frankfurt, Germany, in March of this year. This was another fantastic experience that allowed me to meet other like-minded individuals from different countries and learn about new and innovative technologies in the plumbing industry.

Through my journey in the plumbing industry, I have come to appreciate the importance of hands-on learning and the value of apprenticeships. I strongly believe that apprenticeships are a fantastic way to acquire the necessary skills and knowledge to succeed in today's job market. Being involved in the apprenticeship system provides individuals with a pathway to a successful career and helps to bridge the skills gap that exists in many industries. I am incredibly grateful for the opportunities Jones Engineering and the plumbing industry provided me. My experience with apprenticeship has been nothing but positive and I would highly recommend this route to anyone who is looking to excel in a chosen field. My journey is a testament to the value of apprenticeship and hands-on learning. I am also excited about my plans for the future. One of my primary goals is to move beyond the tools into a higher role in my field. I have gained a lot of valuable experience as a tradesman and I am now ready to take on more responsibility and challenges. In addition, I have a deep passion for travel and exploration and I hope to see more of the world in the coming years. Travelling can be an excellent source of personal and professional growth. I am lucky that, with my profession, I can go anywhere in the world. I am committed to continuing my journey of lifetime learning and upskilling as I believe that staying curious and open to new knowledge is essential for success in any field.

Mr. Martin Scattergood: I am a sheet metal worker and am now fully qualified now. At school, I liked science, technology, engineering and maths and I always preferred to be much more hands-on in my life. When doing the junior certificate examination, I had to concentrate and knuckle down to do well and pass. I passed, but I found it very difficult and had to do a lot of work for it. I knew at an early stage that I wanted to do something with my hands and decided to go down that route. After doing the junior certificate examination, I knew I would struggle

doing the leaving certificate examination. I heard about the leaving certificate applied, LCA, through a friend. He had done it in another school. It is much more hands-on. The school I was in did not allow me to do art, metalwork, woodwork and computers at the same time, but they would have allowed students to do all the science subjects. The timetable in the school did not suit, so I moved school and took the LCA which included art, metalwork, woodwork, computers and many other more hands-on subjects and outdoor activities. This opened doors into an apprenticeship for me. When I left school, I had done all the subjects in school. I had completed projects. I had a lot of worksheets and so on. We had done big projects in school, so I was able to go straight into an apprenticeship. As someone who struggled in a normal academic leaving certificate class, I think students should have different routes available to follow a more practical route in STEM.

Deputy Mairéad Farrell: Gabhaim buíochas leis na finnétithe as ucht teacht os comhar an choiste. I thank the witnesses for coming before the committee. It is my first education committee meeting and it has been quite interesting to hear all the different points of view. I have a few questions. My first question for both witnesses relates to the current housing crisis. We need to get as many people trained in skills through apprenticeships to be able to build the houses we need to deliver for people. One of the things Mr. Bermingham mentioned was how difficult it was for him to get the initial apprenticeship he wanted to do as a motor mechanic and that his family connections meant he was able to go into plumbing. In many ways, like in a lot of things it seems that if you know someone, doors can open for you. We are reliant on the private sector to be able to provide the different positions. How would the witnesses feel about the possibility of it being done through the public sector, for example, through a State construction company that could provide those apprenticeships so that people can get through? It would not exclude this, but rather provide more opportunities.

Mr. Bermingham also mentioned that he did a competition in Germany. My Mam is from Germany so I have a particular interest. Things are done well there. Doing the leaving certificate equivalent to go on to third level is not the only route to do things like nursing. For example, my cousin is a nurse. She never did her leaving certificate examination. She did a training course after school. It is the same for physiotherapists. When Mr. Bermingham was in Germany, did he get much of an understanding about how their apprenticeship models work? Is that something that came across? If not, that is grand. I would be interested.

Mr. Scattergood stated that it was only through the fact that he had a friend who was able to do the LCA that he became aware of it. I went to a small secondary school in Galway. There were subjects that I could not do. I really wanted to study history. I did so for my degree but I could not do so at school because not enough people wanted to do it. The issue of not being able to do all the subjects you want will always be there. Schools will place value on certain subjects depending on the teachers they have and so on. Do the witnesses think there is a need as part of the education process that schools are clearer to students about the different subjects that are available? It was quite courageous for Mr. Scattergood to move school after the junior certificate examination in order to do what he wanted to do, because he knew what he wanted to do. From what I understand, the possibility was only clear to him because he had a friend who had done the LCA. The school did not readily tell him that it was available. I would be interested in the answers to those questions. The witnesses can answer whichever ones they want.

Mr. Jamie Bermingham: The Deputy spoke about the possibility of the State providing an apprenticeship programme to help to get more people through. It is a great idea but it would be important for the quality of the work to ensure that the right people are teaching the apprentices.

I found in working for family businesses, that I was not always getting the training I should have been getting. That can happen even in big companies. I sat on an independent panel with SOLAS recently and a big issue that came up was preparing apprentices for what they will be doing in their phases and having the right people who are willing to take the time out to train them properly. That is how we will get good quality tradespeople.

Deputy Mairéad Farrell: That is interesting. I used to work in banking before I was a Deputy. I worked in two different investment banks. They were big corporations. One of them was far better at upskilling. It was a matter of luck. Can that also be the case in apprenticeships?

Mr. Jamie Bermingham: For sure. That is a big thing.

The second part of the Deputy's question related to German apprenticeships. My Dad brought me into the apprenticeship route because he knew I always wanted to work with my hands. From when I was 14 or 15, he used to talk about how Germany did it. It is done all the way through secondary school. It is drilled into students because they are so big on apprenticeships. That is what we need here.

Deputy Mairéad Farrell: That is interesting.

Mr. Martin Scattergood: Regarding what the Deputy said about the LCA, everyone from my LCA class is doing apprenticeships now or many of them are fully qualified and have moved out of home. They are doing well.

Deputy Mairéad Farrell: They can afford to.

Mr. Martin Scattergood: Yes. I am trying to move out as well.

Deputy Mairéad Farrell: Do you think there is an issue that schools are not giving their pupils the available options? Then we have a situation in which people are struggling through doing something they have zero interest in or aptitude for.

Mr. Martin Scattergood: The school I went to was an academic school. I wanted to finish school but I was going to have a hard time doing so. I wanted to work with my hands. My grandad was a sheet metal worker as well. I was interested in that type of work. Through a friend I went down that route. Changing school was hard.

An Cathaoirleach: I will go back to allow the witnesses who are late to read their opening statements and then we can return to where we were.

Mr. Shane O'Connor: We thank the Chair for inviting us here today and for the opportunity to present our research project Assessing the Impact of Second-Level Education on Key Aspects of Adolescents' Life and Development and to contribute to this very valuable discussion of the progression of STEM in Irish education. The perspective that we can provide is as students ourselves and from our research.

Throughout the past three years of conducting this research, one of the main aims of our study on the impact of secondary school on the experiences of students was for it to translate into actual policies implemented at a grassroots level. To provide some context on the study itself, our research was conducted over the years 2020, 2021 and 2022, focusing on the impact of second level education on students' social development, physical health and mental well-being as well as gauging the impact of remote learning. Undoubtedly, each of these areas influences

a student's learning abilities and the optimisation of the school experience is vital in strengthening the value that students reap from their time in school. Many of our key findings present tangible opportunities for growth and improvement within the education system.

Mental well-being is a topic of discussion in many classrooms across the country. This is evident in the volume of opinions on the matter from both students and teachers. Students who are likely to report a negative impact of secondary school on their mental health are females, senior cycle students and students in single-sex female schools. While it is evident that mental health is something that teachers and students are acutely aware of, 72.8% of students indicate that they do not receive educational guidance on their mental health within the school environment and 33% of teachers believe that their schools do not take action to support students' mental health. Students and teachers most frequently cite practical well-being classes as a possible improvement that could be made. While both parties are aware of the literature available on the topic, they feel a more practical application of this knowledge would be the most productive approach.

Social development was undoubtedly the most positively rated area among students and teachers. Being with friends and socialising were the most frequently cited positive aspects of second level education. While the proportion of students who believed that second level negatively impacted their social development and interaction was in the supreme minority, these students felt that excessive homework and exam pressure were limiting factors.

In terms of physical health, males had a much more positive outlook on the impact of second level on their physical health versus their female peers. In several cases, focus group results showed a severe lack of physical education, PE, classes in many single-sex female schools. This is reflected by the low ratings from these students. Another key area we wish to highlight is the dropout rate in sports. Some 13% of students dropped the majority of their physical activities while in secondary school. Most strikingly, 85.9% of these students dropped their activities during junior cycle, with the majority doing so in first year.

One of the most inspiring results from this research came from a studied mixed DEIS school where simple, informal and short lunch-time physical activities resulted in this group of students moving from the lowest rating in terms of physical health to the second most positive in year 2, with the trend continuing in year 3. Physical health is strongly correlated with social interaction and mental health. Correspondingly, when the students in this school improved in terms of physical health, so too did their social interaction, social development and mental well-being. This displays that simple actions from school management and leaders can often have exponential benefits for the student experience.

I will now pass over to Liam.

Mr. Liam Carew: I thank the committee for inviting us to this meeting. Along with the aforementioned areas, students were also surveyed on the extent to which they felt second level education equipped them with the skills to progress in life. In this question, transition year students rated the lowest. Given that STEM is a skills-heavy field, this indicates a significant opportunity for the transition year programme to be more reflective of the value of STEM in acquiring skills.

Unsurprisingly, Covid-19 negatively impacted all key areas assessed. Students' social interaction and development was most adversely impacted by the pandemic, with results showing a significant decrease since the pandemic's beginning. However, there was a slight rating gap

recovery in year 3 of the study as we moved to more stable times. It should also be noted that physical health was the least impacted area, followed by mental health.

Significant input was received from teachers regarding their concerns over students' lack of development arising from periods of lockdown and online learning. Many reported their students having decreased skill sets, an ability to concentrate and an ability to learn compared to pre-pandemic cohorts. This will become one of the true challenges of our generation, and a challenge for educators and students today will be a challenge for third level institutions and employers tomorrow.

Based on the valuable results received over recent years, we have been able to form several conclusions and recommendations. In terms of mental health, we believe that the best way to foster a good mental health status among the student body is to create positive and caring student-teacher relationships along with providing excellent pastoral care. Furthermore, students are not always aware of the mental health supports available to them, rendering these systems and protocols less effective. A campaign in each school to emphasise the resources available to students would be productive.

Social interaction is a vital aspect of school. This was emphasised by students unequivocally throughout our study. Students should be given every opportunity to socialise and participate in peer-based activities within their schools.

In terms of physical health, the negligence of PE classes by certain schools evidently has impacts for many students. Group sport activities that involve the whole school community contribute to improved student experiences across the board. Students in DEIS and non-DEIS mixed schools should be encouraged to participate in physical activities to a greater extent while students in junior cycle should be encouraged to keep up extracurricular activities and be shown the benefits of staying engaged with their sports.

Without a doubt, students have been impacted heavily by the pandemic, particularly in terms of social interaction and their learning and development skill sets. Students will require support to provide them with practices they missed during the period of online learning as well as equipping them with the skills they need to progress in life.

Our research shows the profound effect that education and schooling have on an adolescent's life both in a negative and a positive way. It also highlights the challenges that have become apparent after prolonged periods of remote learning during Covid-19. It is our aim that every effort be made to enhance the student experience in such a pivotal time in their lives and development.

We thank the committee for its time and look forward to discussing our research in education with it.

Deputy Sorca Clarke: I thank the witnesses for joining us. Mr. Scattergood hit the nail on the head when speaking about the routes through education and to a career. We do not live in a society – if we are honest with ourselves, we have not done so for quite some time – where someone's route through school and college looks like a motorway. In the majority of cases, it looks like a country bóithrín. It takes twists and turns before people eventually find themselves at a place in their lives where they have fulfilling careers. This relates to what Mr. Scattergood said about options at secondary level in terms of subject availability or the leaving certificate applied versus the traditional leaving certificate.

Mr. Scattergood said he came from an academic school. Mr. Bermingham did not mention what his school was like. What was the reaction from their peers, families and teachers when they said they wanted to pursue a trade? Within these four walls, we speak a great deal about third level, but we need to know from people in the trades what they believe would happen if people were encouraged to opt to pursue trades as opposed to going to third level. Clearly, both witnesses are successful in the careers they have chosen for themselves. Looking back on the academic and on-site elements of their apprenticeships, what needs to change? If there were people who dropped out of apprenticeships along the way, what engagement did the witnesses have with them and what reasons did they have for leaving?

Mr. Martin Scattergood: The first school I went to was all about college, college, college. That is what it was geared towards. When the teachers were speaking about that, I thought that I was not going to college and would definitely work with my hands. I could not see myself in college, but I just got on with it. The second school I went to had a much more relaxed atmosphere. The teachers were friendly and geared us up for the workforce, for example, showing us how to write CVs, do interviews, get a job and so on. When I came out of school, I got straight into an apprenticeship. I had to wait two years for there to be enough numbers to run a sheet metalwork class in college on Bolton Street, but it was great when I got in. Funnily, Bolton Street once held a meeting of all the apprentices who were there at the time. It took place in a big meeting room or hall and all the apprentices sat at the back. The head of the college said there is a stigma around being an apprentice. It was not the teachers who made us feel like that but we were in a college full of other students and we were just the apprentices down the back who were not there for a long time. The head of the college said everybody needs to sit at the front. He made us feel very much at home and told us we were just as entitled as everyone else to be there. As apprentices, people tend to keep the head down and get it done. There is a view that we are not really in college but, in fact, we are.

Deputy Sorca Clarke: It is a type of unconscious bias.

Mr. Martin Scattergood: It was not coming from the teachers. I cannot stress enough that the teachers in Bolton Street bent over backwards for us, and I would do the same for them. They made us feel really at home.

There was no one telling me about apprenticeships in junior cycle at school. I had to hunt out the information for myself. We were told about the college route and that was it. As regards STEM subjects, I did technical graphics and maths, as well as engineering in my first year in school. However, the school timetable did not suit and I could not continue with engineering, which is metalwork, because it conflicted with other stuff I wanted to do. I had to choose art instead of engineering. That was more important to me at the time.

Mr. Jamie Bermingham: At my school, I was kind of pushed through it. I did not have much of an interest because I grew up being persistent that I would become a mechanic in the family garage. Individual teachers in my school were very encouraging when I mentioned the idea of an apprenticeship, but the idea did not come from them. It came from my personal experience and growing up with the family garage and around cars. That was where my mind was set and I knew what I wanted. At the same time, there were other teachers for which apprenticeships were not really a thing. It was about going to college because that was the route they took and everyone else was doing the same. There were very few of us who were being a little different and looking to pursue an apprenticeship because it was what we wanted to do. I was always mad for working with my hands. That is how it was for me. It was 50:50 in school with the judgment, with some teachers saying it was a great path to go down and others not.

Deputy Sorca Clarke: Looking back over his training, both on-site and academic, does Mr. Bermingham see any areas in which there needs to be improvement or things that can be done better?

Mr. Jamie Bermingham: More funding would be a big thing. I brought this up at a meeting with SOLAS recently. I did my phase 2 apprenticeship training in Gweedore, Donegal in 2019. One guy dropped out because he was so far from home. That is a big thing at the minute, especially the difficulty of finding accommodation through all the phases. I was lucky enough to be in Cork for phases 4 and 6, where I was able to stay with family. However, some of the lads struggled. There was the extra pressure of being away from home and it was costing them a lot of money. There was a lot of extra stress.

Deputy Sorca Clarke: It should not come down to luck as to where people are sent.

Mr. Jamie Bermingham: Not at all. Those are two big issues that definitely need to be looked at.

Deputy Sorca Clarke: I thank the witnesses.

Senator Fiona O'Loughlin: I apologise for being absent for some of the meeting. I had a Commencement matter for discussion in the House and another meeting. I got the opportunity to meet Mr. Carew and Mr. O'Connor briefly in the Seanad Chamber. They are both very welcome, as are Mr. Bermingham and Mr. Scattergood. It is good to hear the voice of experience and what they can tell us about their discussions with their peers.

The study by Mr. Carew and Mr. O'Connor is important. While we hear a lot about mental health, including positive mental health, and the programmes that are being rolled out and need to be rolled out, it is really important that we hear from a peer-led review. The witnesses referred to practical well-being classes as a possible way of improving mental well-being. Do they have any examples of schools that are providing such classes and the difference they make?

Mr. Shane O'Connor: In our experience, we had activities that were peer-based and based around year bonding. That really brought us together as a year group. We had ours in September and it brought us together for the year. We have seen across schools that when people can rely on a community and they are all tightly knit, it has a very positive impact. Where schools have a social-pastoral system where students mix together positively, as well as students and teachers, it has positive effects on students' mental well-being, and that is correlated to all the other areas of social development and physical health. In our experience in our school, we certainly have seen that positive actions have had positive impacts, and that is also to be seen in other schools.

It is important to reinforce that students and teachers are very aware of the literature around mental health and mental well-being. That was emphasised again and again. They really think that something like going for a walk will make a real, exponential difference to students. We have seen throughout the survey and the study that small actions have massive impacts.

Senator Fiona O'Loughlin: I thank Mr. O'Connor. I have some questions on apprenticeships for Mr. Bermingham and Mr. Scattergood. They spoke about where the impetus came for them to pursue an apprenticeship, which for Mr. Bermingham was to do with his background and the family business. The number of apprenticeships has gone up significantly, as has the number of women participating. Back in 2018, only 2% of apprenticeship places went to females, which was concerning. That percentage has increased, although probably still not by

enough.

It is really important that guidance counsellors give a strong and positive message about apprenticeships and the ability to earn as you learn. It is a very valuable way of entering the workforce. Do the witnesses feel, in their experience, that there is enough emphasis on this from guidance teachers? Once people start an apprenticeship, is enough information given to them on the pathways into full employment?

I am of the view that of the three blocks of work experience students do during transition year, one of them should be focused on an apprenticeship area. Do the witnesses feel more could be done on that during transition year?

Mr. Jamie Bermingham: A lot more could be done in transition year. I did my work experience in a garage, which was a big thing for me. It was kind of the dealbreaker. I knew after fourth year that this was what I was going to do. There definitely should be much more emphasis in school around apprenticeships. Schools should be pushing them more. Doing an apprenticeship is fantastic. It takes four years and you earn while you learn. When you come out, the whole world is in front of you. I qualified in January and I am now sitting back and thinking about what I will do for the foreseeable future. I am looking at going on to do a degree and moving up the ranks in work. It is fantastic. It is not just four years and then you are done. It is a whole lifetime of learning and opportunities. As I said in my opening statement, apprenticeships open numerous doors and opportunities. That is what is so great about them.

Mr. Martin Scattergood: I did the leaving certificate applied, LCA, which includes four weeks of job placements. I did two weeks in an office and two in a garage. After the two weeks in the office, I knew for sure I was going to work with my hands. When I was in school, nothing was said about apprenticeships or about working with your hands, including being a brickie or electrician. That was never mentioned. It was all about going to college and doing science. That is perfectly fine, but the information about apprenticeships never got to us.

Senator Fiona O'Loughlin: I thank the witnesses.

Senator Malcolm Byrne: I thank all of the witnesses for sharing their stories and congratulate them on their successes. I listened to the point on having experience in an office and knowing one definitely did not want to work there. My first summer job was picking strawberries at home in Wexford. That assured me I did not want to have anything to do with horticulture or anything like that.

In respect of apprenticeships, there are huge technological changes. Skills that apprentices are acquiring now may be redundant in five years' time and in ten years' time things may be completely different. Do the apprenticeship models we have prepare people sufficiently to deal with some of those technological changes?

Mr. Jamie Bermingham: I had a meeting with SOLAS recently to review the curriculum for my trade, plumbing, and there was also a group of others in attendance. They discussed the modules. There are phased modules in college, and we discussed what they were teaching and whether it is and will be relevant in the future. Some are and others are not. Phase 2 is the step in.

My major focus is renewable energy, and I feel there should be more focus on renewable energy in phase 2. People study certain subjects in phases 4 and 6 and cannot go forward unless they do a separate course with a different body afterwards. Mixing and matching and adding in

the extra stuff that will be relevant for the future, in particular renewable energy which is very important, should be introduced, especially at the early stage of apprenticeships.

Mr. Martin Scattergood: My grandad did an apprenticeship about 70 years ago with Irish Rail. He was in Bolton Street and I am pretty sure that when I was there some of the same machinery was still there. I am sure his initials were on the desk; my nanny told me to look out for them.

In terms of sheet metal work, we do a lot of machine work for catering equipment and so on. That requires a lot of computer work. When people need something burned or CNC lasered out, companies expect people to know how to use these machines. My company does not have one, but we do a lot of that work. People who go to college are shown how to do the work on a computer screen, but they need to use the equipment. Sometimes the equipment is not available. There is a lot of old machinery which needs to be upgraded.

Companies are sending people to college to learn how to do their trade, but when they get there the trade is about 50 years behind given the equipment that is available. The teachers are fantastic and make things work. People watch a lot of slide shows and stuff like that. Students are using computers. There is software called AutoCAD, for which there is a licence. The college only had ten licences, but there were 12 in our class. We had to switch out the licences on the computer. If we had 20 licenses for the software we would be able to use it, but the teachers made it work. Everything went fine. The teachers make it work, but on the sheet metal and welding side, as well as metalwork, things were somewhat behind in comparison to electricians. There are tonnes of electricians. All of it needs to be upgraded.

Senator Malcolm Byrne: My next question is related to further education and training. From their studies, do the witnesses think the education system values how we learn in a wide variety of ways? Is there too much focus in the education system on the academic side? What is the perception of the witnesses?

Mr. Liam Carew: Aside from moving on to third level after secondary school, part of our research is based on the skills students need to progress in life in general. We posed this question to all sixth years in schools in Ireland and those doing the LCA. Transition year students responded with the lowest rating and felt they were not prepared to move on in life. That surprised us, because students in first to third year come into secondary school and are innocent. They think they are going to learn a lot about how to progress in life. Transition year is the year where people go on work experience and do all of the things that will help them make a final decision come sixth year. Students realise that they have not been taught those things. There is a bit of a gap.

Mr. Shane O'Connor: The rating drop in terms of skills between third year and transition year was definitely a result of people going on work experience and seeing that the skill sets they have get them so far but not all the way. Moving forward, we could have a multitude approach whereby people learn some skills and then apply them. It is when that happens that people learn how little they know and can learn more. Practical as well as theoretical applications can create a strong framework.

Senator Malcolm Byrne: How important are extracurricular activities such as the BT Young Scientist and Technology Exhibition to a student's individual personal development? I ask the witnesses to draw from their personal experiences as much as their research.

Mr. Liam Carew: From personal experience, it was super to be involved in such a competition. We competed online in 2021 and did not get to see the other students' projects. We learn about science and STEM and everything that goes with that in school, but when we go to competitions like the young scientist exhibition in Dublin and see over 1,000 students presenting the work they have put in we get to see science and technology in practice. It is great to see. It helps our development and we learned a lot from that.

Mr. Shane O'Connor: People are applying skills they are learning. This was our second year competing in the competition and it was great to meet people who have similar interests to us and compete against them. The buzz of the whole competition was great. What we have learned since has been even more valuable. Opportunities like today are something we have not had before now and we are very grateful for them. I do not think we would be sitting here today without our involvement in that. It provides a wealth of opportunities.

Deputy Cathal Crowe: I have followed the debate remotely. I am probably the last speaker. I congratulate all of the witnesses. There are some award winners here, some of whom have picked up very prestigious awards.

Mr. O'Connor and Mr. Carew said a lot in their opening statement, but I want to hone in on physical education. When I was in second level, from third year onwards I could count the number of PE lessons we had on one hand because everything was driven by academics at the time. I very much understand that was the bad old days and now subjects like PE are protected, which gives youngsters some headspace during the academic week. I ask the witnesses to elaborate on what they uncovered, because it featured in their report. How prevalent is such a practice?

Mr. Liam Carew: Of the schools we studied, the students who rated their physical health the lowest were females in secondary schools. We conducted our surveys, followed by focus group in the schools we studied. We found that when students reach fifth and sixth year, in particular sixth year and the lead-up to the mock exams and leaving certificate, PE classes can often be abandoned and students are given a study option during those classes. The Deputy said his understanding was that such classes are protected these days, but for students who do not play sports outside of school the two PE classes in the course of the week might be the only physical activity they are doing. When they are taken away and replaced with study classes there is nothing for them throughout the week. We found the classes are being replaced and cast aside as the spare option for extra work to be done.

Mr. Shane O'Connor: PE classes are a vital aspect of school life because they reinforce the idea of a school community. Often, we do not see everybody in our school year because of subject choice and so on. We are all in different places at the one time. PE is an opportunity for students to come together and do something. It adds to the fabric of the school community. The negligence regarding PE in some schools was very concerning because they are not providing students with the same opportunity to mix.

Deputy Cathal Crowe: Is the practice widespread?

Mr. Shane O'Connor: It was widespread. It is most prevalent in single-sex girls schools and mixed non-DEIS schools. We found that a great emphasis put on PE in boys' schools. It can fall through the cracks there as well.

Mr. Liam Carew: There may be a stigma in certain schools about PE classes not being es-

sential or necessary. That feeds down from higher up, namely, from the principal down to the students. As a result, those classes will be neglected in certain schools.

Deputy Cathal Crowe: That is a matter of concern. If possible, it would be good for the committee to write to the inspectorate and the NCCA in order to ensure that maximum hours are still being devoted to PE. I say this because it becomes a pressure valve when people get into the last ten or eleven weeks before their State examinations. The pressure is immense. Everyone is talking about their future and what they need to be studying. People are worrying about what they are not doing and what they are neglecting. An hour of running around and chasing a ball, being in the school gym or whatever is a great pressure valve to just break away from all of that. Other members covered that in their questions, but it is a matter of particular concern for me.

In the time remaining, I want to go to Mr. Bermingham and Mr. Scattergood. I do not want to guess their ages, but they are a few years older than Mr. O'Connor and Mr. Carew. I remember when I was in second level, when people were in sixth year, if they spoke about an apprenticeship, somebody would say to them, "That is great, but why do you not try UL or NUIG first?" Parents were guilty of doing that. Indeed, teachers were also guilty of doing it. Everyone was saying, "Just go this way and you can think about it afterwards, but go off and get your four years on campus first". Do Mr. Bermingham and Mr. Scattergood think that is still a factor? Anecdotally, one hears a lot of youngsters saying that they want to be carpenters, blocklayers or plasterers but their parents will tell them to go up to Dublin for a few years. When they come out of college, they do not want to be hands-on. They want to be the project manager or the foreman. On many construction sites, most of the labourers are more than 40 years old. It is rare to have someone the same age as the witnesses on a site. What would they say to people who are in their leaving certificate years? Is it a factor that people are being lured away from that path and sent a different way? Is that skill set being lost?

Mr. Jamie Bermingham: There is a big stigma in sixth year regarding apprenticeships. They try to change your opinion of them. I do not know if it is just an older opinion, let us say, where it is judged that you will only get so far by going that route. As a matter of fact, it is not true. As I mentioned earlier, if you do your four years, you will have that behind you and then you can go on to do anything else. You can go back to do level 7 or level 8, and you can keep going. That needs to be expressed and shown much more so that people can see that you can go further than just doing a four-year apprenticeship and qualifying as, for example, an electrician, plumber or whatever. Those are the important points that need to be considered and expressed much more - 100%.

Deputy Cathal Crowe: I wrote to the clerk and Cathaoirleach of the committee a few days ago to say that the career guidance teachers associations should come before us to discuss a multitude of issues, including this one. I very much see it in my constituency with Shannon Airport and the huge industrial zone there. It is one of the largest industrial zones outside Dublin, where there is huge amount of foreign direct investment, aircraft mechanics, aircraft sprayers, everything - you name it. Many of these foreign companies, for example, Lufthansa, are making major investments. They have extremely ambitious plans, but they cannot get the workers. They are not coming out of UL, they are not coming out of TUS and they are not coming in at apprentice level. This lack of labour coming through will be the next drain on the economy. Many of those in Mr. Bermingham's age group who qualify will go to Australia or Canada and explore the opportunities there.

I want to conclude by saying to the Cathaoirleach that as we proceed with our work pro-

gramme, the career guidance teachers might come before us at some point.

An Cathaoirleach: I have a couple of questions for the witnesses, the first of which are for Mr. Bermingham and Mr. Scattergood. In their views, what are the biggest challenges that are facing a student or a young person who is considering an apprenticeship? I can see that Mr. Bermingham questioned his confidence going back over the years, but he has gone on and on. I can see how confident they both are now with where they are and with what they both have achieved. In fairness, they have achieved a great deal. I congratulate them on that. It is remarkable. There are many other thousands of students and apprentices who are going through the scheme and who are similar to them. From speaking to fellow students who are pursuing apprenticeships, what do the witnesses believe are challenges?

Mr. Jamie Bermingham: The big issue is being away from home. That is a big challenge. When many people think of an apprenticeship, they know they will be going away and that is a big factor that comes into their heads. I loved it. I enjoy travelling, going to new places and meeting new and different people, so I got on very well with it.

Another big issue is that when you come out of school, you are going straight into college or you are getting job. From experience and from listening to other people, what throws people off apprenticeships is the starting wage, because it is so below the minimum wage. That is a big factor. I have friends who have only recently gone into apprenticeships. They are a little bit older and they struggle because they might have a house, a family and stuff like that. They are really struggling. It is not as though it is a couple of cent under the minimum wage. Some of them are a good couple of euro under. That is a big factor that deters people from choosing apprenticeships, in my opinion.

An Cathaoirleach: That was one of my questions, whether that is a factor. On the issue of getting a sponsor, did Mr. Bermingham find it difficult to get a sponsor or was he helped with that through the college? Did he have his sponsor before he went into the college?

Mr. Jamie Bermingham: Yes. You need to have a company that puts you through the apprenticeship programme and that registers you as an apprentice under the company. The company needs to be interviewed by SOLAS so it is deemed eligible to train apprentices. It gets checked to make sure it has its craft certificates and stuff like that. There are a couple of background checks on employers.

An Cathaoirleach: Does Mr. Bermingham think it is more difficult to get a sponsor in areas like Dublin and in the bigger cities than it is out in the country?

Mr. Jamie Bermingham: I would have said it would be the opposite. I would say it would be easier to get a sponsor in Dublin than it would be out in the country.

Mr. Martin Scattergood: We work for many companies. Probably once a week people would ask me if I have any friends because they need young fellows and they need bodies. There is a lack of apprentices.

I definitely felt in school that it is just about college. College is beaten into you, so to speak. Someone asked if there was a stigma around apprenticeships. There kind of is, but you have to get over that. In schools, there was absolutely no mention of apprenticeships. You had to hunt that out yourself. I did not know about it, only through a friend who told me this course will gear you up for going into an apprenticeship.

An Cathaoirleach: This issue of stigma around apprenticeships has been discussed before here. I asked the two lads about their research and I will come back to them shortly. There is no doubt that there is a stigma. That is my own view and some of the witnesses who have been before the committee would say, and I have used the word, that apprenticeships are not sexy. However, I can say, once a person goes through an apprenticeship, he or she can often earn an awful lot more than people who have done a three- or four-year college course. That has come through here over recent weeks. If the witnesses were to give advice to some of their peers who are 16 or 17 years old, of if they were speaking to a fifth- or sixth-year class where people were considering doing apprenticeships, what advice would they give to them? That might be a difficult question.

Mr. Jamie Bermingham: If I was to sit down with someone who was coming towards the end of their school term and they were considering an apprenticeship, I would have nothing else to say other than just do it, stick at it and work hard, because from my experience, that is what I did and it has got me a long way since. For someone who did not enjoy school so much, my whole life has changed ever since I started an apprenticeship. I cannot value it enough because it was life-changing. It has been and it will be for the rest of my life. Stick at it and work hard. You just have to get through it.

An Cathaoirleach: Before I go to Mr. Scattergood, Mr. Bermingham mentioned that he would like to go travelling. Is that to broaden his career prospects, or is it that he just enjoys travelling and he wants to go travelling?

Mr. Jamie Bermingham: I suppose it is a bit of both. I want to see other parts of the world and to get out of here for a little while. I am mad to go to Canada. I would love to-----

An Cathaoirleach: Would you come back again? After travelling, would you like to return to Ireland? Will you see how you get on first?

Mr. Jamie Bermingham: Ireland plays a big part in my heart, so I do not think I could settle down anywhere else. If I went travelling, I would definitely come home. That is for sure.

An Cathaoirleach: To add to that question, I congratulate all the people who have been involved in WorldSkills, who have been before the committee before. They do fantastic work in getting that competition up and running. It is fantastic for anybody to have it on his or her CV, and the witnesses have acknowledged that. Once you have it on your CV, and getting to Germany and so on, it makes a huge difference to any job prospects.

Mr. Jamie Bermingham: For sure. It is a huge CV enhancement, and not only that, it has changed my whole perspective on the apprenticeship programme since all this has happened. Ever since I won the WorldSkills Ireland award, everything has changed, between how I am looked at in work, who knows me, where I have gone and what has come up. It has just opened so many different opportunities. It is fantastic. As I said earlier, it is very important to have the right training from day one and to stick with the apprenticeship. If you do, that is where you will end up. You will end up at the WorldSkills awards with a silver medal like me.

An Cathaoirleach: Fair play to you.

Mr. Martin Scattergood: The Cathaoirleach asked about any advice to give young people. Just find something you are interested in and go with it and you might get a job in it. Even if you do the apprenticeship in that trade, you still have a cert under your belt and you can do something else. Also, the national skills event used to be done in the colleges. Mine was the

first year it was done in the RDS. That was just one brilliant show, with all the trades under one roof, and all the colleges could come and take part. From doing it just in a classroom to doing it in the RDS was brilliant.

An Cathaoirleach: I have just two questions for Mr. O'Connor and Mr. Carew. First, what inspired them to do the research they did? Second, on something the other lads raised, do they think, as students in second level education now, that there is a stigma around apprenticeships, and what advice would they give to young students considering apprenticeships, as the other lads said, to do one? If you want to work with your hands, there are opportunities out there. I remember, when I was Minister of State in the Department of Defence, trying to get people to do apprenticeships in the Air Corps. There were fantastic opportunities but it was just difficult to get the numbers in.

Mr. Shane O'Connor: The reputation of the young scientist exhibition follows itself through always. It is something that from first year we had the intention of doing, and then the freedom of transition year, TY, allowed us to pursue it thoroughly. That was one of the main reasons we decided to do it. It has been something we have been working on for the past three years. Once we got going and found out that we were quite good at this, we were able to keep going, and that kept us motivated and it keeps you to account. The young scientist exhibition was something that had always been on the horizon and that we had looked for since first year. In our experience - in my experience anyway - the tide is turning with apprenticeships, as the lads said. Many of our peers are looking towards apprenticeships as a viable approach, often more viable than other approaches, so, in my experience, apprenticeships are not stigmatised but are very much recognised for their real value.

Mr. Liam Carew: To answer the Cathaoirleach's first question about our inspiration for entering the competition, we began our research in 2020, when we had just finished third year and when schools were closed in March 2020. Then we entered the competition in January 2021. We were coming to the end of our third year and we had the option of going into TY, straight into fifth year or into the leaving certificate applied. We wondered, when we looked at our year group, what are the factors that impact our peers and our own decisions at this stage and what is the true impact school has on us as students at this age? That was one thing that inspired us to undertake this study.

In terms of apprenticeships, in our school we have quite a small year group of about 50 but there are maybe ten or so students who are looking into apprenticeships at the moment.

Career guidance in school and so on is still geared a lot more towards college courses. Every week representatives come from colleges giving talks to the students and advertising their colleges. Maybe the same is not quite there for apprenticeships. When we hear people in our year saying they want to do this apprenticeship or that apprenticeship, it comes from them and they ignite it themselves. A few weeks ago our school took two groups on a workshop to Irish Rail in Dublin, for example, so there are initiatives definitely coming around, and students are becoming a lot more aware that apprenticeships are available, but I think it still comes from the students who want to do it themselves.

An Cathaoirleach: I thank Mr. Bermingham, Mr. Scattergood, Mr. O'Connor and Mr. Carew for appearing before the committee. The voice of young people must be centre stage in all decisions taken in the education system, and this committee has made a virtue of bringing in people as young as 15 and 16 years of age to have their voices heard. It is very important. I congratulate Mr. Bermingham and Mr. Scattergood from WorldSkills Ireland and Mr.

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O'Connor and Mr. Carew, winners of the young scientist award. Apprenticeships, the young scientist exhibition and WorldSkills Ireland are so important. Mr. Scattergood talked about having the honour of going to the RDS to see other people in the competition and what they do in their areas. I hope all the witnesses have very bright futures. They are all a tribute to their tutors and their families. Their coming before us is very much appreciated.

The joint committee adjourned at 1.46 p.m. until 11 a.m. on Tuesday, 25 April 2023.